

Chapter 2-3 Examples:

1. Assume the following Cobb-Douglas aggregate production function

$$GDP = zK^{\alpha}N^{\beta}$$

Where K stands for the aggregate capital stock and N is aggregate employment.

Assume further that

$$\beta = 0.3$$

Ceteris paribus, if employment (N) rises by 2 percent, what is the growth rate of GDP?

2. Assume labor income tax revenue equal to

$$t \cdot w \cdot N$$

Where t is the average income tax rate, w is the real per-hour wage, and N is the number of hours worked. Assume further that

$$t = 0.10$$

The government hopes that by cutting the tax rate that it can encourage people to work enough more hours that it can earn just as much after the tax cut as before.

- A. Assuming that the real wage is unaffected by the tax cut, how much would N have to grow in order for the government to break even on the tax cut if the new tax rate were lowered to 9 rather than 10 percent? Show your work.
- B. How would you assess whether or not such a change in N is either predicted or plausible?
- C. Is there anything about a tax on labor input that makes you think about this problem differently than if we were looking at the market for pizza? Explain.